Stakeholder-centered Process Implementation: Assessing S-BPM Tool Support

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ABSTRACT
Since Subject-oriented Business Process Management (S-BPM) models can be executed after validation without further transformation, tools have been developed to support model execution. As these tools target not only non-disruptive execution of process models but also intuitive ease of use, stakeholders could expect effective and efficient implementation support of business processes. In the presented study we have challenged 3 stakeholder-centered tools refining, validating, and executing a complex process model. We wanted to know how much effort needs to be spent when a prototypical application is generated from a process specification provided by the involved stakeholders. The tools were a commercially available suite, a tool currently approaching the market, and a research tool. Our assessment study reveals substantial effort that needs to be spent for refining and validating process models, before being able to generate an interactive process experience. Hence, (S-)BPM tool developers are encouraged to support stakeholders according to the identified needs.

CCS Concepts
• **Information systems→Information systems applications**
• **Human-centered Computing→Collaborative and social computing.**

Keywords
Subject orientation, Business Process Management, Workflow Management, process execution, rapid prototyping

1. INTRODUCTION
Digitalization of the economic sector, both, in production and service industry, has (re-)shifted the focus of organizational development towards Business Process Management (BPM) [1]. Stakeholders get increasingly qualified to model and further develop their work procedures in a self-contained way, in particular (re-)designing or adapting business processes (cf. [2],[3],[4]). Since Subject-oriented BPM (S-BPM) claims intuitive modeling and non-disruptive execution of process models, stakeholders could expect from S-BPM tools, not only being able to specify processes from their perspective, but also to refine them to executable work procedures in an effective and efficient way (cf. [5]).

Since the introduction of Subject-oriented Business Process Management (S-BPM) [6] several support tools have been developed either by researchers or software developers. These developments follow different implementation strategies, both for stakeholder interaction, and execution. For instance, StrICT ([7], www.strict-solutions.com) is focusing on Microsoft technologies and thus, the Windows Workflow Model, whereas the S-BPM interpreter developed by Lerchner et al. [8] is based on Abstract State Machines (see also [9]) and Java technology. Although direct execution of models has been provided by those tools for various stakeholders, to our knowledge no study has been published so far assessing the effectiveness and efficiency of tool support for transforming stakeholder-generated S-BPM models to respective artefacts.

Once we can identify the tradeoff between stakeholder process specifications and their execution, S-BPM tools can be improved or redeveloped according to stakeholder needs. In this paper, we challenge a selection of S-BPM tools with different development backgrounds regarding their support for implementing stakeholder process specifications. We want to know, how much development effort needs to be spent to generate prototypical applications from a given process specification. We have selected a commercially available suite (www.metasonic.de), a tool currently approaching the market (actnconnect.de), and an open research tool (http://www.i2pm.net/interest-groups/ueber-flow/home).

We have explored their capabilities implementing a stakeholder-defined process on Project-Based Learning (PBL). This process has been modeled by stakeholders in the course of designing the organization and operation of the School of Education at an Austrian university. The organizational development team of that School designed a curriculum and a corresponding PBL process, involving learners, facilitators, and other stakeholders. The novel concept addresses and involves all stages of education, adjusting capacity building on preschool, junior and high school, academic and occupational education level. Hence, the goal of our study is to identify the effort for refining and validating the curricular process model, thus being able to generate an interactive user experience activating a process execution engine.

In the following, we first recall approaches to BPM tools supporting not only the execution of process models but also the evaluation of engineering performance, before we introduce the procedure we followed in our assessment (section 2). We also

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describe the PBL process that we used as baseline for assessment in this section. After giving a short overview of the S-BPM engines involved in the assessment study in section 3, we report on the results of transforming the stakeholder process specification into executable S-BPM models utilizing the systems under test in section 4. We conclude the paper reflecting on the study, and providing an outlook for future work in section 5.

2. STUDY DESIGN

In this section, we review current inputs on how to design assessment studies of this kind, referring to integrated BPM tools in the first subsection. We then provide the design of our assessment study in the second subsection, before introducing the stakeholder process specification we have used as a baseline for assessment.

2.1 Towards Integrated BPM Tool Support

Although the need for finding the right tool for the different stakeholder needs in BPM has been recognized quite early (cf. [10]), and user-centered modeling has already become a research issue (cf. [11]), evaluation studies target execution from a processing (cf. [12]) rather than a stakeholder perspective. The following review reveals the processing/execution focus, however, exemplifying integrated architectures supporting the transformation of models and simulation of processes.

Conceptual work on evaluating business-process execution engines has started recently by Ferme et al. [13]. They have developed a framework, aiming to handle complex behaviors more effectively and efficiently. The authors have studied the high expressiveness of currently used modeling and execution languages, which in turn results in high complexity of the execution behaviors. Their model-driven approach starts with a BPMN model and a set of Workflow Management System configurations, while allowing the instantiation of an adequate test infrastructure. The framework has been tested in practical use cases.

Gavialis et al. [14] have started to integrate modeling and execution for simulation in the realm of benchmarking. Their web-based system integrates evaluation, monitoring, simulation and redesigning processes. It enables efficient, personalized and consistent services to citizens and businesses at reduced operational cost. A Process Designer component is used for process modeling and performance of simulation scenarios executed in the Process Engine. The Process Analytical Reporting component provides the necessary process performance reports, whereas the Process Monitoring and Exception Handling module deals with real time information about the processes' execution. Those components are integrated with the existing IT infrastructure, including databases keeping business rules and objects.

This BPM System architecture is also integrated with the Benchmarking system. In this way, performance measurement and benchmarking across and within domain systems can be performed. In the presented case municipalities have been the baseline, and processes have been tested based on a dedicated performance evaluation algorithm. Performance indicators were directly linked to the business processes in the overall system. Functionality for process/service analysis, design and redesign could be provided in an integrated way: 'The process models prepared can be dynamically executed with the embedded simulation tool and the simulation results can be fed back to the Benchmarking system. In that way, the results of an in vitro new process process/service execution can be tested and the impact on performance can be evaluated.' (ibid., p. 486).

With respect to BPMN workflow engines Skouradaki et al. [15] introduced the first benchmark that specifically targets Business Process Model and Notation 2.0 (BPMN 2.0), as the addressed workflow engines had to be compliant with BPMN 2.0. They identified as main challenges in benchmarking: '(a) Collecting real-world process models; (b) Synthesizing the benchmark workload out of real-world processes; (c) Designing the benchmark environment; (d) Assessing and selecting the BPMN 2.0 engines to be tested; (e) Characterizing workloads of different actors; (f) Defining expressive Key Performance Indicators (KPIs)' (ibid, p. 301).

They followed an iterative project management approach for designing and releasing the benchmark, in order to enhance both the completeness and real-world representativeness of the benchmark successively. The initial test was a performance stress test with a workload including all elements of the BPMN 2.0 Core3 (to simulate real-world behaviors). Thereby, throughput (i.e. processes executed/time unit) was used as KPI for the selected execution engines. The second iteration involved additional systems (open source and proprietary ones). It included load, soak, and spike tests in addition the performance stress test. In addition, simple actor workload models were used. The workload mix was also more complex in terms of structures, parallelism, and interaction with external services, and BPMN 2.0 non-core activities. Finally, the set of KPIs was enriched by latency and utilization.

The third iteration extended not only the sample of workflow engines, but also the complexity of interactions of the actors and the workload mix, utilizing the complete BPMN 2.0 set. The impact of monitoring to the engine's performance was also assessed. Additional, customized KPIs can be used throughout this iteration. Overall, this procedure can be used for structuring assessment and benchmarking execution support of specified process models.

2.2 Assessment Study Design

In our assessment study, we followed (i) the stakeholder perspective when generating executable process models, (ii) coupling model representations with execution engines for implementing stakeholder processes, and (iii) a process model generated in the field, including all concerned stakeholders and the complexity required to generate a prototypical run for each stakeholder perspective. The design of our assessment study followed the steps (a) – (f), developed by Skouradaki et al. [15] and described in the previous subsection:

(a) The baseline process specification comprises the behavior of each stakeholder involved in studying in a Problem-Based Learning (PBL) process at a School of Education when applying for, achieving acceptance, running, and completing a learner project. Its specification requires several swim lanes due to coordination requirements when modeled in a subject-oriented way.

(b) The synthesis of the assessment workload is based on the behavior models of each subject. Hence, the principle workload is directed by subject orientation, separating the behavior and the interaction required for task accomplishment in dedicated diagrams. The workload is determined by executing instances of each subject according to the flow of messages between them.

(c) The design of the assessment environment is, on the one hand, given by the scenarios encoded in the process models, and on the other hand determined by the evaluation setting. Each tool is tested by a selected master student with 4 years of formal education in Business Information Systems at Johannes Kepler University. Due to that background the students are also familiar with the PBL-based education, and the parameters of concern, namely the effectiveness and efficiency of the transformation process when implementing specified business processes. Hence, the selection of the test persons was intentional due to the assessment tasks in terms of the addressed domains, namely academic studies and BPM. Each of the students underwent an evaluative interview whether the assessment could be done on the same level of competence with respect to BPM, BPMN, S-BPM, XML, tool handling, and transforming models to executable workflow specifications.

(d) The selection of execution engines is based on the idea to involve systems from different backgrounds, and with different degrees of maturity: the Metasonic® suite has been on the market for several years (since 2006), and has been continuously developed according to methodological and customer needs. Actnconnect® is a startup implementing subject orientation in a novel way since 2015, and UeberFlow is a recently published research pilot, publicly available, enabling the execution of subject-oriented models. It is also based on recent technologies.

(e) The task of each study participant, in order to bridge the gap between stakeholder process representations, subject-oriented process definitions, and executable code, comprises a variety of activities: (i) getting familiar with the stakeholder process & prepare the runtime environment for execution, (ii) transforming the stakeholder process model into an executable specification including validation, (iii) generating and executing workflow prototypes, (iv) providing timely entries into the development diary, (v) demonstrating results, and finally (vi) assessing the transformation process & reflecting on the achieved results. This list of activities has led to the milestone plan given in Table 2 in the next section.

(f) Defining expressive KPIs. The most important objective of the study was to demonstrate an accurate runtime behavior based on the stakeholder model input. Hence, runtime performance was one of the KPIs. Secondly, the accuracy of supporting the model transformations was of interest, as it affects the effectiveness and efficiency of implementing stakeholder process models.

Due to qualitative nature of the objective of the study we were interested in the activities the participants of the study actually needed to perform. Consequently, we designed a diary structure for documenting each step. Its structure is given in Table 1. We asked the participants to fill in for each activity, when it occurred, its name, the time and cognitive effort they spent to complete the activity. In addition, we were interested in any detail the participants considered worth reporting with respect to the effectiveness and efficiency of the perceived task support. Specific guidelines to fill in the dairy were handed over to the participants to ensure consistent use of relevant information items, such as time spent for an activity.

Table 1. Project diary structure

Date	Activity	Effort	Details
……	……….	……	………….
……	……….	…….	………….

Finally, we decided that each participant should demonstrate the running prototype another participant had developed. Using a different execution engine than the one used for transformation and presenting the result of another participant, we anticipated, could bring new insights, both, with respect to the effectiveness of transformation process, and the accuracy of execution results.

2.3 Baseline Process

The baseline process stems from a project targeting the design of a newly founded School of Education at an Austrian university. The project team has decided to represent the business logic in a subject-oriented way utilizing BPMN (www.bpmn.org). The School of Education involves many different stakeholders, stemming from different disciplines and organizational perspectives, as it aims to tackle the entire spectrum of educators, ranging from preschool age to professional occupation. The set of stakeholders includes learners, facilitators, registration office, study program board, facility management etc. Such a school and overarching study program has not been implemented so far at an Austrian university. In order to structure the organization and to ensure overall acceptance, the steering committee of the School project required a stakeholder perspective. The authors were invited to follow a role-based and communication-oriented approach when representing the business logic of the school.

From an educational perspective, the implementation of the study program is driven by Project-Based Learning (PBL). This didactic approach is learner-centered, and a multi-facet endeavor: 'In project-based learning, students gain important knowledge, skills, and dispositions by investigating open-ended questions to "make meaning" that they transmit in purposeful way.'([16], p.5) Typical questions are 'Can a complex crisis be handled by involved stakeholders?' and 'What makes 'experts' experts?' Consequently, learners need and learn to design the process of learning, being the core experience in PBL. Hereby, 'questions activate, arousing curiosity and driving students to inquire.' (*ibid*, p.6)

From an institutional perspective, most important is that 'Projects **are** the curriculum – not an add-on – and through them, students develop important capabilities.' (*ibid*, p.6) Meaning is created by the unique experience through a project that should be conveyed to others as it sustains in the mind of the students, regardless whether the project objective was inventing new things, entertainment, persuading others, motivating a topic, or an inspiration. As a consequence, when representing the processes in a workshop with the program heads, the process team followed the PBL design procedure:

1. Identification of project-worthy concepts
2. Exploration of their significance and relevance
3. Identifying real-life contexts
4. Engagement of critical thinking
5. Developing a project sketch
6. Planning the set up including peers

In addition, several institutional goals, such as assigning ECTS points, needed to be considered in the course of process design. Each learner is able and encouraged to submit a project proposal to structure the study program according to individual needs and preferences. Depending on the design and the topic, the proposal attracts peers volunteering for various roles, including the project manager. Once a learner has achieved a basic set up and a first project plan, the ECTS Board and Study Director clear the proposal for further studies, and also provide a coach for that project. In this way, both formal and educational objectives can be met.

The original, subject-oriented representation has been developed with education experts using BPMN. Each stakeholder role or bundles of activities, corresponding either to a stakeholder role, an organizational unit or technical system has been detailed in a swim lane. This fine-grained differentiation of role behavior in terms of over 20 swim lanes reflects the density of required interactions when implementing PBL in an educational setting.

Putting the interaction to practice on such a level of detail, is an effective way to keep the involved stakeholders up-to-date informed and trigger relevant tasks to be accomplished at the right time, even when only a small set of persons is actually involved. As it turned out in the course of modeling, the swim lane identifiers, i.e., the subjects, could not be specified in a straightforward way. They needed to be developed according to needs in communication and exchange of data (business objects) between actors (subjects), i.e. when becoming crucial for successful operation (cf. [17]).

Without going into further detail, the process for each student of the School comprises three phases: (i) preparing a complete project proposal, including attracting facilitators and peers to join or contribute, (ii) getting approval to start by relevant authorities, (iii) running and completing the project, including the provision of results as a common information base. The turnaround time for a project is a study year (winter and summer term in Austria). Results of successfully completed projects become part of the School's organizational memory system. It can be accessed by the public, and thus, provides the basis for novel proposals and continuously developing ideas and concepts.

3. TOOLS

In this section, we briefly describe the tools that we had selected for the assessment study. According to the goal of our study, to find out the effort for refining, validating, and implementing the curricular process model, we identified those tools available that are able to generate an interactive user experience when activating the process execution engine.

3.1 Metasonic® Suite

The Metasonic® Suite (www.metasonic.de) has been developed to support the S-BPM lifecycle (cf. [18]). Once developers follow this lifecycle, they can validate and execute processes whenever an S-BPM model has been created or modified. For model creation, several support tools have been developed. They capture tangible model creation [19], semi-tangible construction of models ([20]; www.metasonic.de/touch), and intangible model generation via a touch-based tablet application [21], or a virtual world setting [22]. Most of these systems can be coupled with the suite, in order to process created models. In addition, the suite provides a dedicated S-BPM editor for modeling and validation.

When using the Metasonic® Suite, BPM stakeholders are supposed to follow three steps to develop a complete business application. In step 1 each process participant models his/her process. Users also need to specify the services needed to accomplish interactive tasks. Each model needs to be validated before proceeding with the next step. In step 2 the created process models need to be linked with IT applications implementing the functional elements of S-BPM behavior diagrams. Then, the organizational implementation needs to be completed. Finally, in step 3 stakeholders are able to execute their process models according to the represented business logic. The suite provide a proper workflow user interface to control the execution.

In order to support each of the previously mentioned steps, dedicated software components are available: *Base* is required for administration. *Build* comprises all design components to describe processes. *Proof* supports validation that is required before execution, and *Flow* is the runtime system supporting the execution of models.

3.2 ActnConnect®

In order to overcome experienced deficiencies in S-BPM projects, in particular related to coupling subjects, business rules, implementation services, and user interaction, Strecker et al. [23] have proposed a dedicated actor concept and developed a corresponding software platform termed 'actorsphere' (www.actnconnect.de). It supports the execution of Business-Actors and Service-Actors and their self-induced choreography. The platform was developed as a Java-based OSGI platform including a web-based user interface. Each actor is represented by a single XML-file, in order to enable independent runtime behavior as well as to manage different versions of actors by using standard (code-)version control systems like GIT. Business-Actors and the originally defined S-BPM subjects differ in a variety of ways:

(a) Business Actors are loosely coupled or operate through different communication channels - subjects are linked together using the same communication channels.

(b) Business Actors cannot be refined like subjects leading to semi-structured operations in subjects, having do-states executed by a machine mixed with do-states executed by a human being. If a Business-Actor is implemented as a Service-Actor, its service layer(s) must provide a full implementation or service mapping for each do-state.

(c) Business-Actors, in contrast to subjects, offer the possibility to define the data representation (= user interfaces) for human users.

3.3 UeberFlow

This S-BPM engine [24] is based on the Actor Model of Concurrent Computation [25]. Any workflow model comprises of a set of actions or tasks, which are ordered in a certain sequence and performed by workflow participants having certain roles. Any Workflow Specification comprise the actors WorkflowUnits, WorkflowSteps, and WorkflowFunctions. The Workflow Specification represents an entirely executable model of the workflow. It acts as container for the WorkflowUnits which group process steps according to the responsible role similar to lanes in BPMN or subjects in subject-oriented workflow specifications.

For each role participating in the specified workflow a WorkflowUnit is created which contains and supervises all WorkflowSteps the corresponding role is responsible for. Additionally, WorkflowUnit functions as a data space for the underlying WorkflowSteps and WorkflowFunctions. In the course of execution all data accessible by the associated role are made available to the WorkflowSteps via the WorkflowUnit.

WorkflowSteps represent the activities of a workflow. The actual execution logic of a WorkflowStep, its prerequisites and results, are solely defined by its WorkflowFunctions. Each WorkflowStep contains a sequence of WorkflowFunctions which are executed sequentially, when the corresponding step is triggered. WorkflowFunctions are the most fine-grained units of execution in the UeberFlow Lang meta-model, and the only active component from the workflow's perspective. Each

WorkflowFunction represents an atomic action of workflow execution. Since WorkflowFunctions encapsulate all actions of a Workflow Specification, different types of WorkflowFunctions are needed for runtime purposes. In the basic version of UeberFlow Lang six WorkflowFunction-types are used.

For each component of the UeberFlow Lang an Actor has been implemented, making use of the Akka framework (http://akka.io/). Consequently, an instance in UeberFlow is equivalent to all actor instances created in the context of this particular workflow instance. All of those actor instances are aggregated using the actor structuring and supervision mechanisms by defining a root actor representing the entire instance.

4. ASSESSMENT

In this section, we report on how we implemented the study. We present the results of each implementation process and tool in separate subsections. We do that in a structured way following the milestone plan given in table 2. According to the milestone plan and the structure of BPM assessments including the execution of processes, we finally can cross-check the results. Based on identified commonalties and distinct differences, both with respect to the transformation process, and the resulting prototype, we are able to argue on the effectiveness and efficiency of S-BPM process implementation support.

4.1 Study Procedure

The study started in February 2016 with preparing the overall setting, and has been performed throughout the Austrian academic spring term 2016 at JKU between March and June 2016. It involved 3 students (one female, two males) - we term them test developers in the following - in Business Information Systems at University of Linz attending a master program course in Advanced Business Engineering, and tool experts to support the students operating their respective tool (environment). The course is rewarding 3 ECTS which means, each of the students should spend around 60 work hours to complete the course.

Table 2. Milestone Plan

Date	Milestone
9.3.2016	M1: Process is familiar, tool is configured and installed
13.4.2016	M2: Tool-conform model is created
11.5.2016	M3: Process can be executed
15.6.2016	M4: Results can be presented, and prototype can be demonstrated

The preparation included contacting the tool producer(s) and ensuring the availability of the software and service support for each study participant. Then, the test developers were introduced to the study objectives and design. The data collection procedure included continuous documentation in form of the diary structured as shown in table 1, regular meetings and online advice by domain and tool experts when needed.

Interesting was the effort spent for (i) tool installation and configuration (M1), (ii) creating tool- and domain-conform process representation (including possible process adaptations) (M2), (iii) implementation and testing to demonstrate the validated showcase (M3), and (iv) briefing another developer when preparing a presentation of results and prototype (M4). In addition, for M3 the participants were asked to identify a part of the test case revealing high value added, and a sample part of the test case that is connected with high risk, according to their understanding of the process.

The documents the test developers were asked to deliver by the end of the term in June 2016 included: (i) adapted and executable process specification, (ii) configuration manual for the tool they have been using, (iii) documentation of the test runs they have been performing, including the results. All test developers were able to complete the study in time and to demonstrate the desired implementation results - the last milestone was accompanied by a presentation to the development team of the School of Education.

4.2 Metasonic®

Since this tool is on the market quite a while, its functionality is documented quite well, and it can be learned in a straightforward way. So we did not expect major difficulties with respect to set up a proper environment for interactive specification, testing, and execution of process models.

Getting familiar with the stakeholder process & configure and set up execution engine (milestone 1). According to the diary, the test developer experienced no difficulties configuring the tool and developing thorough understanding of the PBL process of the School of Education. The suite has been packed in a zip-file and the test developer could easily unpack it and start the included .exe file using a standard configuration on a laptop. The set up could be described in 5 consecutive steps that could be easily followed by a process stakeholder.

Creating a tool-conform process representation (milestone 2). As the suite is ready for modeling immediately after installation, the test developer was able to start modeling without additional efforts. According to Metasonic's 3-layer concept (process overview on the top level, the subject layer containing Subject Interaction Diagram (SIDs) on the intermediate layer, and the internal behavior layer containing Subject Behavior Diagrams (SBDs) on the bottom level), the test developer could map the lane identifiers to subject names and the interactions between the lanes to messages exchanged between subjects, establishing a SID.

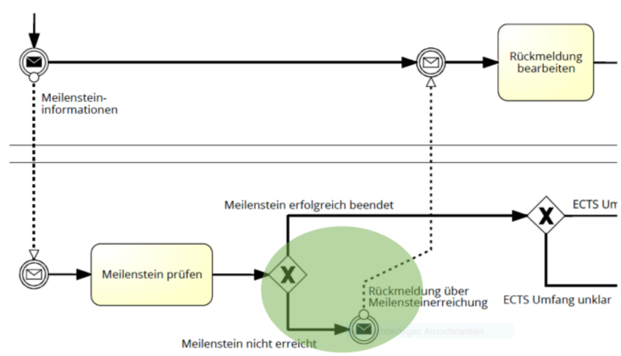

Figure 1. Identification of critical situation

Each swim lane specification corresponds to an internal subject behavior and was represented in a SBD. According to the original specification of the School's operation, 3 different process groups were defined: (i) the annual event for kicking of the PBL procedures – students are invited to participate in the event, (ii) approval process including the acceptance and check of project proposals, (iii) application procedure for project submissions. These 3 process groups could be executed using the suite's Navigator after corresponding business objects (data) were specified conform to the messages that needed to be exchanged.

In the course of modeling, the test developer could identify a part of a process that could lead to critical situations, e.g., misled messages. In the case shown in Figure 1 a milestone achievement is addressed. The activities in the rectangle are ‘check milestone’ (Meilenstein prüfen) and ‘work on feedback’ (Rückmeldung bearbeiten). As indicated by the green circle, a critical situation occurs when the project coach (lower part) has checked the successful achievement of a milestone, and the project leader (upper part) does not a receive feedback on that. As specified in that diagram, only in case of rejection by the project coach feedback is provided. This deficiency has been corrected in the course of validation.

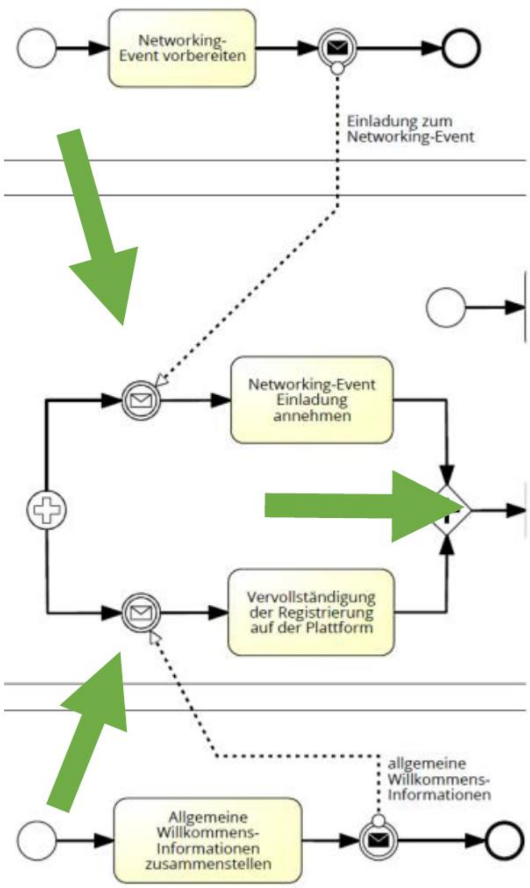

Figure 2. Two subjects start a (Re)Searcher instance

Achieving a running prototype (milestone 3). When mapping the adjusted models to the execution tool of the suite, not only a license had to be ordered from the company, but also runtime problems occurred. Figure 2 reveals the case that 2 subjects start the subject behavior of (Re)Searcher, i.e. the subject triggering the PBL process – see large arrows in the figure pointing to the critical part’s incoming messages and synchronization for further activities. The activities to be performed in this process step are to prepare event information by the event organizers (top lane), accept the networking event invitations and complete registration information on the platform (middle lane = (Re)Searcher), and prepare welcome package by study program management (bottom lane). This situation could not be handled on the modeling layer – it required intervention during runtime. The tool does not allow several start subjects. Overall, several validation problems had to be tackled due to semantic deficiencies, such as

- (re)searcher can send acknowledgement to project management on readiness to participate although project management could already have closed this process
- learn diary for students not specified w.r.t to content structure, which needed to be refined
- program management may not receive required information from project coach or domain coach which in turn could lead to lack of information when checking the project plan
- project leader could send messages about milestones to project coach without guarantee of receiving an answer at all.

However, the milestone could be reached in time.

As the value added provided by the process groups the test developer identified the capability to locate current operations for each stakeholder involved - the representation model allows capturing the state of affairs from a role, functional, and data processing perspective. From the domain point of view, such an expressive representation ensures that a project idea does not get lost, and completed projects are preserved for further use.

The risks related to the process was recognized in the complexity of the model, as stakeholders need to fully understand the process when adapting to another business logic or operational constraints. Albeit the completeness of interaction for the situation at hand, changing a small part of the model might trigger modifications of interactions. One way to overcome this difficulty in maintaining the process is to proceed in small steps and execute them for interactive validation.

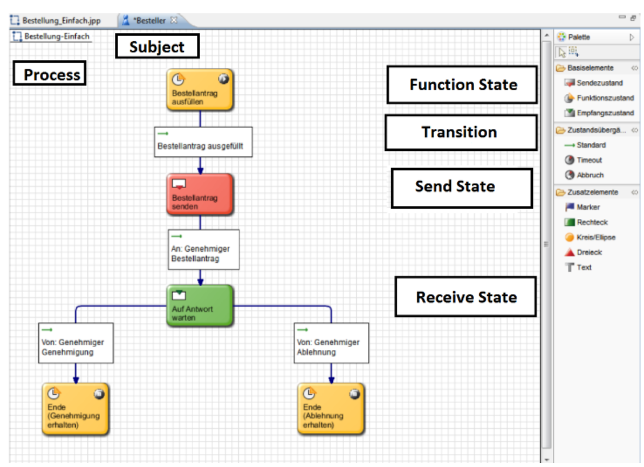

Figure 3. Example of a subject behavior diagram

Presenting the results (milestone 4). Figure 3 and 4 show part of the presentation of the transformed models (subject behavior in Figure 3, and running prototype in Figure 4), as provided by the test developer and presented by another study participant. In the course of preparing the presentation the test developer re-designed the business objects. Executing the models and running the prototype in the suite did not reveal any problems after adjusting the models and the runtime operation as described for milestone 3. The demo showed for the project team as value added the transparency of running a complex business operation. It also triggered reflecting on the business logic in terms of the possible risk of lacking to meet all formal and technical requirements when running the School of Education.

The introduction of the prototype took several hours, adding up to 50 hours of overall work effort for the test developer.

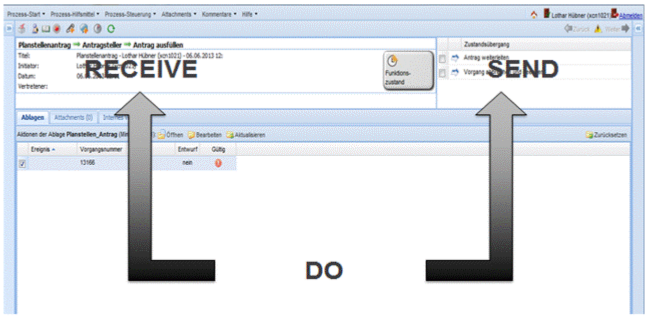

Figure 4. Example of an implemented process step

4.3 Actnconnect®

As already mentioned, this tool is still under development for market entry via cloud or on-premise service provision. The modeling tool allows implementing processes in an actor-based way exchanging messages. Actors represent roles that can be assigned to one or more users in order to run processes interactively.

Getting familiar with the stakeholder process & configure and set up execution engine (milestone 1). The tool is web-based and can be utilized by any browser and an Internet connection. The actorsphere (www.actorsphere.de) activates the user environment and access data provision. The entry screen shows available actors and assigned instances for the current user (Figure 5).

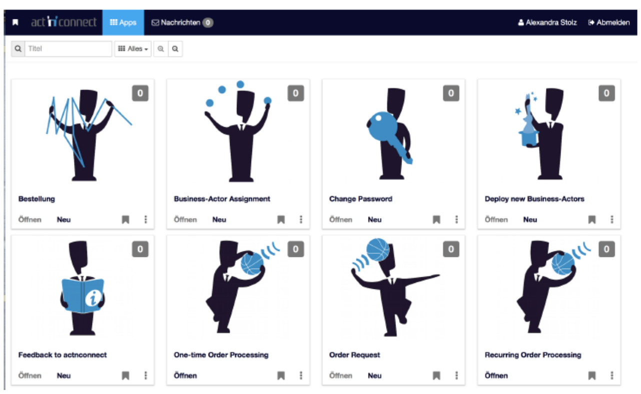

Figure 5. Entering an Actorsphere

The start page contains three relevant actors for each user per default, namely Change Password, Deploy New Business Actors, and Business Actors Assignment. The latter are required to load newly generated actors into the platform, and to assign them after successful deployment, respectively. Actors can be updated dynamically. In case a message exchange is affected, messages of the concerned processes are updated once the corresponding actor instance is restarted. The test developer could manage both tasks, getting an understanding and configuring the tool environment in time. However, it required operational support, as the documentation turned out to be incomplete at that time.

Creating a tool-conform process representation (milestone 2). Each actor is represented by a single XML file that can be manipulated by any text editor. The basic structure is shown in Figure 6 for ‘do’ and ‘send’ operations. Metadata tags are required for deployment. Behavior tags can contain several behavior representations or states (send, receive, do). Each data structure needs to be specified in a dedicated part.

```xml
<behaviors default="standard">
<behavior id="standard" name="Standard" start-state="erstelle_idee">
  <do id="erstelle_idee" name="Erstelle die Projektidee">
    <data>
      <field readonly="false" ref="nameIdee" />
      <field readonly="false" ref="beschreibungIdee" />
      <field readonly="true" ref="baInstanceId" />
    </data>
    <transitions>
      <transition id="send_to_PM" name="Idee senden"
        next-state="send_to_PM" />
    </transitions>
  </do>
  <send actor="portfoliomanagement_SFU" id="send_to_PM"
    message="Send_Projektidee" name="Projektidee senden" partner-role=
portfoliomanagement">
    <mapping>
      <data message-ref="nameIdee" ref="nameIdee" />
      <data message-ref="beschreibungIdee" ref="beschreibungIdee" />
    </mapping>
    <transition id="receive_from_PM" name="receive_answer_bidding"
      next-state="receive_from_PM" />
  </send>
```

Figure 6. XML Structure of an actor (do and send)

The test developer specified the stakeholder process according to the given XML structure and the semantics of the basic elements. Each swim lane of the School of Education process has been initially implemented by a single Business Actor.

Achieving a running prototype (milestone 3). In order to display data, they need to be represented in the respective state using the data tags, and require presentation variables (text fields or graphic symbols) at the user interface. The user interface (UI) is a complex structure, since each element needs to be defined in an XML structure. The test developer utilized a sample UI provided by the tool developers. Problems occurred due to the complexity of some of the swim lanes, in particular the project leader, Curriculum Committee, and (Re-)Searcher. In addition, it turned out, once the UI of an Actor is activated, an instance of that Actor is activated.

Rules are effective means to implement checks according to the state of a process, ranging from user inputs, process properties, to specific requirements actors need to meet, such as instantiating an actor by 2 messages without predefining their sequence. The test developer utilized rules in the trial to implement to check user inputs, e.g., to mark UI fields as mandatory input fields.

Deploying Business Actors starts with selecting and uploading specific XML files (see Figure 7). Based on a syntax check (see Figure 8) the Actor is assigned to a user or a user group. An Actor

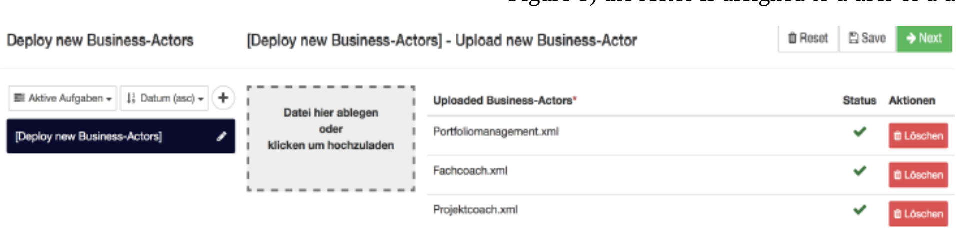

Figure 7. Actor deployment step 1

that can be started is marked with New. Once a message is received, the Actor symbol is marked accordingly with increasing a counter on the top right. Then, the message can be read and tasks can be accomplished as shown in Figure 9 for articulating a project idea. ‘SFU_Lehrbetrieb Projektidee’ means ’project idea to be handled in the School of Education’s curriculum’. ‘Erstelle Projektidee’ means ‘create a project idea’.

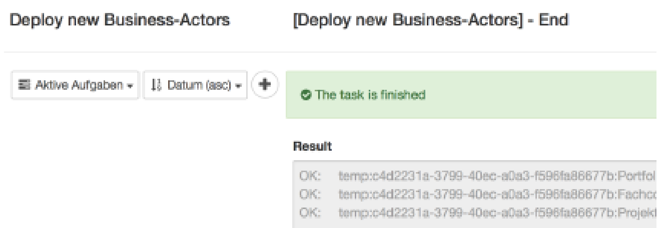

Figure 8. Actor deployment step 2

When executing the Actors as initially specified it turned out that the assignment swim lane to single Actor should be re-designed in a task-oriented way, e.g., splitting the behavior of (Re-)Searcher to critical interaction, getting informed and receiving approvals for the project. The most critical part was identified for the project leaders, as project planning might involve several iterations before a project might actually start. For each critical interface, e.g., between Project Coach and Curriculum Board, dedicated solutions have been specified in the original swim lane model before being implemented at the Actor level.

Figure 9. Started actor for having a project idea

Presenting the results (milestone 4). Once several runtime problem could be resolved, in particular, instantiating an Actor requiring more than a single message in arbitrary sequence, most of stakeholder processes could be executed and demonstrated interactively (Figure 9). The interaction in the course of presentation did not reveal any problem. Overall, the test developer spent 80 hours to implement the stakeholder process, mainly due to re-arranging Actors and addressing the adjustment of UI functionality with the execution of Actors.

4.4 UeberFlow

The tool focuses on execution of processes and was originally built to implement a meta-model for communication- and function-oriented representations [24].

Getting familiar with the stakeholder process & configure and set up execution engine (milestone 1). In order to run UeberFlow, several applications, namely IntelliJ IDEA, Node.js, MongoDB, and JDK 8, are required. After adapting paths to run UeberFlow some setup activities, e.g., starting Node Server and MongoDB, a corresponding UeberFlow project for implementing a process model could be started by the test developer. In contrast to understanding the stakeholder process, this process caused several iterations and was guided by the tool developers.

Creating a tool-conform process representation (milestone 2). The stakeholder process has been implemented step-by-step. In order to handle the increasing complexity the test developer introduced a process structure plan. It contains each process steps providing (i) name of process step as given by stakeholders, (ii) main processes, (iii) sub processes, and, (iv) data to be sent. The restructuring and XML-based implementation revealed as value added of the process that a project is never implemented without sufficient resources.

Achieving a running prototype (milestone 3). In order to generate a process model in UeberFlow, a ProcessModelFactory should be specified. Each Process Model consists of Process Units containing Process Steps, Metadata, and ExecInfo, such as Roles. In the course of implementation, the test developer experienced difficulties sending variables, as it may require Require, Proceed, or Provide Functions. In order to facilitate the dynamic evolvement of process designs, a Git repository was installed. When executing the process, several critical parts have become evident, among them the irreversibility of a (Re-)Searcher decision to participate in a certain project, unless the project has been completed.

Presenting the results with another engine (milestone 4). The tool supports a dashboard containing all running processes, and to run activities within instances of processes. Several adaptations were implemented, e.g., masking help variables, in order to optimize runtime behavior, adding up to an overall effort of 87 hours of the test developer. The presentation has been performed by another test developer without major hindrances after a short preparation. Figure 10 gives an impression how in UeberFlow a student (‘Projektideenträger’) delivers a project idea.

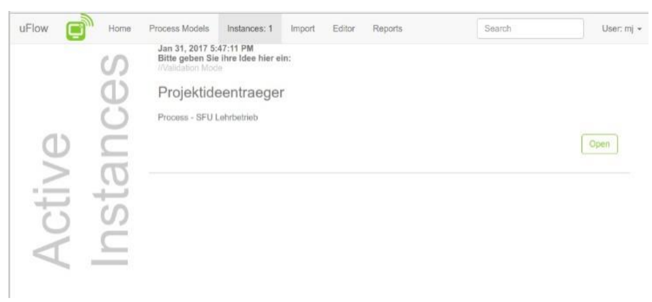

Figure 10. Sending a request

4.5 Summative Assessment

From a general perspective the data show that the most convenient support can be provided by the tool already on the market, whereas the others required (manual) XML-based transformations for executing process models (see Table 3). This effort also became visible when setting up the implementation environment, UeberFlow requiring effort due the various components to be installed, ActnConnect causing effort due to the lack of documentation by the time of the assessment (see also Figure 11). UI adaptations are required for both, as theirs focus is on runtime aspects rather than generating interactive user experience.

With respect to the redesign efforts of the original stakeholder process specification, the implementation required some restructuring effort (transformation effort), in order to manage the complexity of the process, and thus, specifying the proper granularity of subjects: In case of Metasonic 3 process groups, in case of Actnconnect Actor decomposition based on the swim lanes, and in case of UeberFlow an orthogonal rearrangement in form of a Project Plan had to be performed. All cases indicate to better bootstrap process design, immediately starting with toolsupported prototyping, albeit having to recognize tool-specific conventions or adaptations of the process.

Looking closer and clustering the data of the study, as shown in Figure 11, it turns out that all 3 systems performed well with respect to runtime support, once all interferences on the instance level had been removed. Looking at the congruence of the date of introduction with the support capabilities, the tools were experienced as congruent. Hence, the tool available for the longest time period, could meet corresponding expectations, whereas the 'newcomers' did not induce high expectations. The same holds for the ease of installation and configuration. Regarding the latter, developers should take care not to build hindrances, in order to get a system running, in particular, as the overall usability has been experienced satisfactory and beyond.

Table 3. Summarizing Experiences

Development activity	Metasonic® Suite	Actn-connect	UeberFlow
Configuration	No major effort	Support required	Individual Set Up
Process Specification & Validation	Build Process grouping	XML Actor grouping	XML Process Models
Execution	GUI template	GUI template	Instance Manager GUI

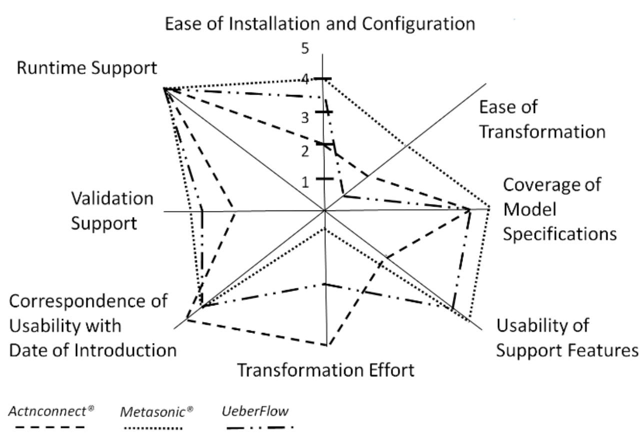

Figure 11. Clustered assessment data

Most of the development work needs to be spent on transformation support, in particular recognizing difficulties on the instance level while modeling or validating (ease of transformation). Critical parts of process designs should be recognized at an early stage, in order to avoid several iterations in specification.

5. CONCLUSION

Although tools provided for (Subject-oriented) Business Process Management need to focus on functional needs when transforming models to executable specifications, additional support seems to be valuable. In particular, in case process participants operating their business start working with process modeling tools and execution engines, and thereby, are transforming models to generate workflow applications. These stakeholders could expect effective and efficient support to accomplish the implementation task from S-BPM tools.

In this study we have assessed the effort informed stakeholders need to spend for refining and validating until being able to execute business process models designed by domain experts. We have included into the study three existing S-BPM tools, namely a commercially available suite, a ready-for-the market tool, and a research tool. Although each of the tools support interactive execution of process models, they have different backgrounds, including entrance to the market. In contrast to traditional benchmarking focusing on workflow capabilities, we have explored tool capabilities to transform and execute a stakeholder-defined process model. It represents Project-Based Learning (PBL) that evolved from developing a School of Education.

The study participants were asked to identify and document their effort when implementing the stakeholder process model, starting with the initial stakeholder model and generate an executable specification running on one of the selected process execution engines. The results reveal the current tradeoff between stakeholder understanding and executable models. According to our findings, even S-BPM informed stakeholders need to spend substantial effort refining and validating process models, before they are able to generate an interactive experience for operating a certain process. Key is the validation, both from a functional task and the message exchange perspective, as an implemented subject should be able to act in parallel and synchronized, in order to accomplish an overarching business task, in our case triggering and completing a student project.

For tool developers, several areas of concern could be identified. One relates to the effort setting up a working transformation and implementation environment (efficiency). This should be an easy-to-accomplish and straightforward task. Of crucial importance for implementation is the meaningful support of the transformation task from stakeholder-centered models to executable specifications (effectiveness). The interplay of runtime issues and design time aspects need to be carefully studied for each type of S-SPM execution engine, and should result in providing more intelligent transformation features for process implementation. From a user perspective it is not so much a usability but rather a semantic problem *beyond* the application domain, since critical situations in operation occur at the interface of instances and model representations. Empirical findings from the field, in terms of usability, methodology, and functionality, as e.g., provided by Oppl et al. [26] could guide future research to that respect.

For assessment study designers this research was open with respect to evaluation dimensions, and resulted in first attributes and perspectives concerning the accuracy of support of the implementation process itself, and the implementations that can be achieved by utilizing a certain tool or implementation approach. Although these attributes can be visualized along a spider diagram together with actual values concerning the assessed tools, further methodological inputs should be provided for follow-up studies of this kind. In this way, a set of parameters concerning effectiveness and efficiency can evolve and consolidated from qualitative field studies.

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